## Both 63-bit targets recovered on one RTX 4090

The smaller explicit bases completed the entire ten-execution V21 panel: eight 52-bit controls and two 63-bit prime-field targets. Every scalar passed a fresh CPU certificate replay and the scalar+1 negative control. This is a completed prescribed rung, not a measurement of the GPU's global maximum.

| 63-bit seed | Audit total (s) | Cold solve (s) | Relations | Target queries |
|---|---|---|---|---|
| 101 | 2600.976385 | 2600.961379 | 338 | 15,737 |
| 202 | 1192.904874 | 1192.887598 | 344 | 13,733 |

Measured audit totals include fresh replay and negative-control execution. The relation oracle accounts for2600.586033/1192.429319 seconds respectively; linear algebra takes40.328/41.349 milliseconds. These L0 shared-host clocks are descriptive (wall_valid=false); no controlled timing advantage is claimed.

### Same-service 52-bit controls

| Seed | b: 2R+2 | u: ceil(1.5R)+1 | v: ceil(1.4R)+1 | x: repeat v |
|---|---|---|---|---|
| 101 | 3.044364 | 4.542738 | 6.634252 | 6.626250 |
| 202 | 2.723673 | 4.689136 | 4.806434 | 4.829474 |

Cold solve seconds from rows.csv. Every arm uses twelve slots and R=ceil(r^(1/12)). Smaller bases reduce relation counts from251/261 (b) to 185/192 (v), but increase queries from341/306 to5947/4008 and increase the observed cold time. Same-threshold v/x certificates are byte-identical; recovered scalars agree across thresholds. Order is b/u/v/x then x/v/u/b.

All ten recorded executions are plotted individually from results-v21/rows.csv. The two63-bit points are armv; intermediate sizes53–62 were not prescribed.

## Memory admission, validation and provenance

| 63-bit seed | Left rows | Complete right-cache rows | Admission bytes |
|---|---|---|---|
| 101 | 259,713,792 | 786,806,784 | 16,008,818,600 |
| 202 | 549,596,520 | 444,762,500 | 14,920,341,256 |

Admission is a conservative allocation bound, not measured peak VRAM. Both large cells fit their complete right cache. Partial-cache fallback was checked separately with forced-fallback fixtures; partial allocations remain charged even when full caching wins. V19′s recorded memory/work refusals remain unchanged; V20′s preflight compilation failure produced no experiment cells.

Hardware: one NVIDIA GeForce RTX4090, 24,564MiB, driver595.91.07, compute capability8.9; CUDA13.2 / sm_89 and Rust1.88.0. RunPod pod g2naamm7l8d25t in US-NC-1, provider rate USD0.89/hour. Controller, bootstrap and validation all exited0. The complete archive was downloaded and hashed; pod deletion and absence from a fresh full provider inventory were verified.

Command: bash research/explicit_relations_e2e_20261030/run-memory.sh. Public prime-order fixtures, seeds101/202, p and r below2^63. The driver checks primality, curve membership and the unique Hasse-interval order witness. Target coordinate lifts supply no logarithm to the reduction. All arms use 128 times (columns+1) target queries per trial, at most16 trials. The existing API default stays20. Caps: rows2^32, work2^34, chunk2^20, columns16384, reserve25%; solves180s/7200s at52/63bits, fresh replay60s, fixture180s, worker18000s/controller18300s. These explicit thresholds do not assume the paper's sufficient L>=10m condition.

All29 Rust tests, host/device differential checks, Clippy, nine query variants under each of memcheck/racecheck/synccheck (27 checks), service certificate checks and cap headers passed before data. The exact10-cell native coverage audit passes its fault-injection controls. Schema and file hashes pass. The canonical rectangular checker preserves five findings:16 assumed rows versus10, b/u/x cover one size, and arm instance sets differ. No planned native cell is missing or excluded. Canonical findings remain in canonical-check.log.

Cold solve includes input generation, startup, validation, base construction, all relation attempts, elimination, extraction, internal replay, serialization and teardown. Build/preflight and shared curve generation are outside it. CPU user/system time and peak VRAM were not measured. Periodic archive uploads add host I/O and do not restore solver state. No external-tool timing comparison was prescribed; this is a configuration/admission and recovery observation.

Source: 253abb737139e0e0eaac49bc2bb8a296f353f67c, clean source archive. Protocol SHA256: e15407b5acb04f557cfb0476dbe8ae00a4973583f6bf7be3e7435962d125174b Archive SHA256: 08b89e95da968f85bdf4f7d8c1f8d789cb9bc69d912083258fcf8219db65f50a Raw records were pushed as ee2a1cb4c613 before this report. Full certificates and traces are in validation/ecbench-explicit-x/e2e-20261030-v21/results.tgz in the approved artifact bucket; archive-manifest.json hashes every member. Literature and threshold rationale: SMALLER-BASES.md and FULL-RIGHT-CACHE.md. Next: specialize affine subtraction in the dominant query path; the separate NEXT-AFFINE-QUERY.md records EFD formulas and required exceptional cases. That candidate is not yet implemented or measured.